

Assignment 2 — Data Structures

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Veterinary Office

a. Pseudocode

I would make a patient structure that keeps the important information about each animal.

Patient

name

species

breed

age

ownerName

ownerPhone

medicalHistory

medications

allergies

notes

b. Required, optional, mutable, and non-mutable variables

I think the **name**, **species**, **ownerName**, and **ownerPhone** should be required. This information is needed to identify the patient and contact the owner.

Things like **breed**, **age**, **medical history**, **medications**, **allergies**, and **notes** could be optional because some patients may not have information for every category.

Some variables will eventually need to be changed. Things like **age, medications, medical history, and notes** can change when the animal gets older or receives new treatment. I would not want the main identification information to be changed accidentally. Though things such as an ID should not be able to be changed.

c. Waiting room

For the waiting room, I would want a data structure that makes it easy to add a patient when they check in and remove them when they are called. I would also want the patients to stay in order so that the patient who has been waiting for the longest is normally seen first. The most important things I would look for are easy insertion, easy removal, and keeping the patients in the correct order.

d. Emergency patients

The office has to start seeing emergency patients; therefore, I will need to alter the way in which I build this project. The reason is that an emergency patient needs medical help sooner than a non-emergency patient. Therefore, I am looking for a data structure with a built-in method (or methods) to assign a priority to each patient, as well as to locate the next patient to receive medical attention based on their assigned priority. In addition, if there were multiple patients with the same priority level, I also want them to be treated in the order in which they arrive at the clinic.

2. AI Image Classification

a. Design considerations and bottlenecks

One key thing I would have to keep track of in this project is the 100 images with their respective labels. In my case, I will use 70 images as training data and 30 as test data. In every single epoch I am going to go through every one of the 70 training images only once. Also, I want to assure you that the corresponding image and its label remain together. i.e. A picture of a cat should always be labeled as "cat".

b. Selecting and using data structures

I would look for a data structure that makes it easy to store and go through all the training images. I would keep the training images separate from the testing images, so they do not accidentally get mixed together. I would also keep each image with its label. For example:

TrainingData

(image1, cat)
(image2, dog)
(image70, bird)

TestingData

(image1, dog)
(image32, penguin)

I would want the data structure to make it easy to access each image and its label. The process something going along the lines of:

FOR each epoch

FOR each training image

get image

get label

train the AI

END FOR

END FOR

FOR each testing image

get image

get label

test the AI

END FOR

This would make sure that every training image is used once during each epoch.

c. Shuffling the images

If I needed to shuffle the training images, I would keep the image and label together. I would shuffle the pairs instead of shuffling the images and labels separately.

```
TrainingData = [  
    (image1, cat),  
    (image2, dog),  
    (image3, fish)  
]
```

```
shuffle TrainingData
```

This makes sure that the label stays with the correct image. If I shuffled the images and labels separately, an image of a cat could accidentally get the label "dog." Keeping them together prevents this problem.